

Unilateral Versus Bilateral Exercise and the Role of the Bilateral Force Deficit

Ramsey M. Nijem, MS and Andrew J. Galpin, PhD

Department of Kinesiology, Center for Sport Performance, California State University, Fullerton, California

ABSTRACT

EXERCISE SELECTION IS A MAJOR DETERMINANT OF TRAINING ADAPTATIONS. FOR EXAMPLE, SQUATTING WITH BOTH LEGS (BILATERAL [BIL]) PRODUCES DIFFERENT RESULTS THAN SQUATTING WITH 1 LEG (UNILATERAL [UNI]). SOME SUGGEST THAT UNI EXERCISES ARE SUPERIOR BECAUSE OF THE BILATERAL FORCE DEFICIT (BLFD): AN OCCURRENCE WHERE THE SUMMATION OF FORCE FOR 2 UNI EXERCISES IS GREATER THAN THE FORCE OF A SINGLE BIL EXERCISE. HOWEVER, OTHERS QUESTION THE RELEVANCE OF SUCH A PHENOMENON. THE PURPOSE OF THIS ARTICLE IS TO BRIEFLY REVIEW THE BLFD AND COMPARE BIL AND UNI EXERCISE TRAINING SO THAT PRACTITIONERS CAN MAKE THE MOST INFORMED CHOICE WHEN DETERMINING EXERCISE SELECTION.

INTRODUCTION

An enormous relationship exists between chronic resistance exercise and both sport performance and general muscle health. Understanding the specific adaptations to various forms of exercise is therefore vital to all. Exercise choice is one of

the fundamental modifiable variables and massively influences adaptations (20). This is probably why great debate persists over the proposed superiority of some exercises. A popular example is the comparison between 2-limb or “bilateral” (BIL) and single-limb or “unilateral” (UNI) exercises (2,33). BIL exercises require simultaneous contraction of the same muscles in contralateral limbs (e.g., squat, deadlift, bench press, etc.). UNI exercises restrict contraction to 1 limb individually (e.g., rear foot elevated split squats, single-arm dumbbell bench press, etc.).

Some argue that BIL training is superior because it enables greater absolute force production (i.e., a 2-leg press would produce more total force than a 1-leg press) (2). However, the existence of a bilateral force deficit (BLFD) challenges this argument. This phenomenon suggests that the total amount of force produced during 2 UNI contractions is greater than the force produced during a single BIL contraction (2,13,21–24,28) (Figure). In this case, the sum of the force produced during a right and left single-leg press would be greater than the force produced during a single-effort 2-leg press.

The BLFD seems to be exclusive to homologous contralateral muscle contractions (e.g., left and right

legs) (14) and has been documented in numerous populations (2,12–14,18,21–24,28,34,36,39,40), yet not in others (8,10,11,14,15). The need for UNI force production during chronic human ambulation is conceivably a cause of the BLFD (6), but this claim lacks verification. In fact, several studies actually report a BIL facilitation; meaning the force produced bilaterally is greater than the summed UNI force (9,14,35). This general ambiguity makes outlining the physiological mechanisms responsible for the BLFD difficult. Numerous reports indicate that the issue is primarily neurological (1,14,39), but this position is challengeable (2,4,15).

Knowledge about the influence of both BIL and UNI training on the BLFD is also conflicting (17,37,38). Limited data show BIL training to reduce the BLFD, whereas UNI training tends to increase it. However, perhaps of most surprise is the dramatic lack of evidence linking the BLFD to actual physical performance (e.g., jump height, running speed, squat strength, etc.). It is unknown if such a phenomenon hinders, helps, or is

KEY WORDS:

bilateral; unilateral; bilateral force deficit; functional; symmetry; squat; force; muscle activation

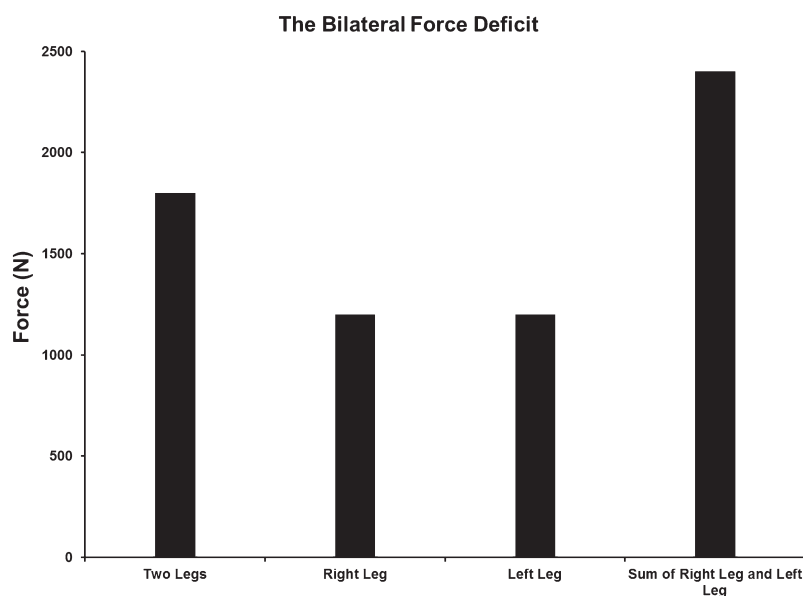


Figure. The bilateral force deficit, demonstrated theoretically by the summed force (column 4) of each leg individually (unilaterally) in columns 2 and 3 being greater than the force produced with 2 legs (bilaterally) in column 1. Columns are numbered left to right.

completely irrelevant to physical ability. Information is available regarding the effectiveness of short- and long-term UNI and BIL training (27,29). Nonetheless, great confusion exists surrounding the BLFD and UNI/BIL training, as a comprehensive discussion of the available scientific knowledge does not exist. Therefore, the purpose of this article is to briefly review (a) the evidence of the BLFD, (b) the potential mechanisms of the BLFD, (c) how the BLFD responds to chronic training, and (d) how chronic UNI and BIL influence actual performance.

EVIDENCE OF THE BILATERAL FORCE DEFICIT

The BLFD was first introduced scientifically in 1961 (12) and suggests that the maximal amount of voluntary force exerted by a single limb decreases when 2 identical limbs contract at the same time (2). Initial work on the topic revealed that 1-leg jump height was ~58% of 2-leg jump height, suggesting a BIL jumping

deficit of nearly 20% (6,41). Koh et al. (19) found similar results with BIL contractions producing ~17–25% less torque and ~20% slower rate of torque generation than UNI contractions. The BLFD has since been reported in a variety of exercise movements and intensities in young, adult, old, active, nonactive, athletic, motor-disordered, male, and female populations (2,12,13,18,21–24,28,34,36,40). Interestingly, the magnitude of the BLFD is clearly contingent on the circumstances of the given situation (e.g., exercise selection, intensity, participant training status, etc.).

For example, leg presses yield a smaller BLFD when performed isometrically compared with at high speed (40). Hernandez et al. (13) observed the relevance of age and exercise intensity when they examined UNI and BIL elbow flexion at a range of intensities (25, 50, 75, and 100% of maximal voluntary contraction) in a group of old (73 years) and young (22 years) participants. Age did not influence the BLFD as no differences

were evident between the 2 groups. This general conclusion has since been supported (23,24). However, the study did find a more pronounced BLFD during submaximal contractions (~14–18%) when compared with the maximal condition (~11%). This contrasts the findings of others who used a similar protocol in the lower-body musculature (knee extensions) of young athletic males and found that a significant BLFD was only present during the maximal condition (21).

Discrepancies even exist regarding the presence of a BLFD as Jakobi and Cafarelli (15) examined young untrained males and actually found no differences in total force, rate of force development, motor unit firing rate, or activation of the leg musculature at any exercise intensity during isometric knee extensions. A BLFD was also absent during knee extensions in a cohort of young (30 years), middle-aged (50 years), and old (70 years) men (11). Moreover, a BIL facilitation was reported elsewhere in a group of men and women of the same age groups (10). These data collectively suggest that the BLFD may be condition specific and that although age may not be a factor, participant training history and exercise velocity, intensity, and choice likely are. A 2001 review highlighted this final point by concluding that a BLFD is frequently expressed during the leg press exercise, but only occasionally during the knee extension exercise (16). Therefore, not only does the BLFD exist, it generally seems to be more prevalent in untrained individuals (14) (though not always (34)) and when engaging in multijoint exercises (16,17). This has led many to believe that limitations of the central nervous system underlie the BLFD.

POTENTIAL MECHANISMS OF THE BILATERAL FORCE DEFICIT

Neurological factors are indeed the most scientifically meritorious of all the plausible causes of the BLFD (1,14,39,40). These explanations rely on the assumptions that (a) neural

activity actually differs between UNI and BIL movements and (b) that difference is large enough to significantly reduce performance during BIL activities. A central neurological relationship clearly exists between homologous limbs as demonstrated by an increase in maximal voluntary force in the left leg after electromyostimulation of the right leg (14). Division of attention does not seem to be a limiting factor as cocontraction of nonhomologous muscles (e.g., arm and leg) does not produce a BLFD (14). Although this basic connection is relatively unquestioned, evidence of differential neural activity between BIL and UNI movements is conflicting.

For example, indirect signs of selective recruitment of fast motor units during UNI movements are supported by some (19,40), but refuted by others (13,23,24,30,31). More direct measures of neurological activity also show equivocal conclusions. When a BLFD is present, a difference in muscle activation during UNI and BIL exercises has been advocated in some investigations (18,39,40), but not others (4,15,22,34). Jakobi and Cafarelli (15) used superimposed tetanic stimulation to assess the ability to voluntarily activate a muscle and found no difference during UNI and BIL leg extensions. It is important to note, however, that this study also reported no BLFD. Thus, their data do not support or deny any causation, but simply suggest that when a BLFD is not present, neural activity is equal between UNI and BIL activities. In contrast, a separate study provided strong support for this neural deficit hypothesis as they noted a general correspondence between reduced neural drive, force production, rate of force production, and muscle activation during BIL exercises in the hand and leg (39). They also used superimposed tetanic stimulation and found that maximal voluntary activation was significantly lower during the BIL trials.

The cause of this reduction in neural drive is unknown. It is conceivable

that an enhanced need for the activation of nonprimary and/or stabilizer muscles amplifies the BLFD as it appears most prevalent during multi-joint movements. This conclusion is speculative, however, as activation of these muscles is usually not measured, and all available data are conflicting (22,26,39). Magnus and Farthing (26) found a link between exercise complexity, stabilizer muscle activation, and the magnitude of the BLFD when they analyzed activation of the rectus abdominis and external obliques during UNI and BIL hand and leg exercises. Opposing conclusions are evident in other studies (39). Thus when taken together, the available literature indicates that neural drive likely plays a role in the BLFD (14,39,40), but the specifics of this relationship are unknown (4,22,34).

HOW CHRONIC TRAINING INFLUENCES THE BLFD

Cross-sectional research suggests that sport and exercise training history influence the prevalence of the BLFD (14,28,35). National caliber rowers (Danish) have a less-pronounced BLFD than their club-level counterparts, whereas international rowers have no BLFD whatsoever (35). Likewise, Howard and Enoka (14) found a BLFD in sedentary individuals, but not in weightlifters with BIL training experience or competitive cyclists. In fact, these weightlifters and the abovementioned international rowers (35) actually displayed a BIL facilitation. These data suggest that chronic high intensity and/or volume BIL training decreases, eliminates, or even reverses the BLFD. The same is presumable for absolute strength levels, but this lacks direct verification.

Available longitudinal studies indicate that training alters the BLFD as short-term (6 weeks) BIL training has reduced the BLFD equally in young/middle-aged (18–35 years) and old (55–75 years) men and women (23). However, the BLFD may respond differently to chronic BIL or UNI training

(17,23,37). A long-term training study (26 weeks, 3 times per week) asked middle-aged (~57 years) women to engage in resistance exercise (leg press, knee extension, hamstring curl, lat pull-down, bicep curls, shoulder press, and chest press) in either a BIL or UNI fashion (17). Only the BIL group reduced their BLFD, prompting the authors to conclude that BIL exercises are preferential when training for activities that require BIL contractions (e.g., 2-legged jumping, standing from a seated position, etc.). Further, the change in the BLFD was inversely related with an individual's initial BLFD, implying that those with a greater BLFD more easily reduce their BLFD with BIL training. A similar short-term study found that UNI training not only failed to eliminate the BLFD but also tended to increase it. Impressively, the investigators repeated the experiment 3 times (once each in the hands, arms, and legs) with the same general conclusions (37). Interestingly, the paucity of data suggests that the effects of BIL training on the BLFD are not body part specific (37), but are dependent on movement specificity (23) (e.g., extension versus flexion). These training studies seem to tell a fairly cohesive story, but more work is needed to confirm these inferences. Most importantly, none of these studies speak to the relationship (or lack thereof) between the BLFD and actual physical performance (e.g., jump height, etc.).

A recent publication examined elite sprinters and reported that the magnitude of the BLFD during a jumping test was inversely related to sprint start performance as well as total impulse of force and velocity on the starting blocks (3). In other words, those who displayed the largest BLFD were slower off the starting blocks. These findings provide indirect rationale for the inclusion of BIL exercise in a sprinter's training program. Obviously, this must be interpreted with tremendous caution as outside of this single article, the relationship between the BLFD and whole muscle performance is

completely unknown. It is entirely possible that situations exist in which having a BLFD is completely irrelevant or even advantageous, but these are currently undocumented. Therefore, until a clear relationship is established, the effectiveness of UNI or BIL exercises should not be contingent on how they influence the BLFD, but by how they influence actual UNI and BIL muscle performance.

HOW CHRONIC UNI AND BIL TRAINING INFLUENCE MUSCLE PERFORMANCE

The concept of training specificity suggests that at minimum, UNI training improves performance during a UNI task, and likewise for a BIL task. Leg endurance seems to follow this model as UNI training does not improve endurance during a BIL task (and vice versa), even though both legs are trained (32). In contrast, training style may not be as important for the development of strength and power as BIL and UNI training equally improve BIL and UNI performance (17,29,32). This was highlighted in a recent short-term (8 weeks, 2 times per week) training study involving untrained men and women. BIL and UNI training caused similar improvements in most testing batteries (BIL vertical jump, 5 repetition maximum UNI and BIL squat, and the Margaria-Kalamen stair climb test); with the exception of the UNI vertical jump, which improved the most in the UNI group (29).

This information is valuable, but also suffers limitations as the long-term effectiveness of training programs may not be fully distinguishable until several months or more of implementation. A clever study addressed this issue by measuring performance changes in a group of untrained, but physically active, women experienced in explosive activities. Testing (BIL and UNI vertical jump, a 10-second Wingate, and a 5 jump alternate leg bound test) occurred after 6 and 12 weeks of either BIL or UNI plyometric training and after 4 weeks of detraining (27). The UNI group improved significantly in all markers

by the midtraining testing. However, no further improvements were made from midtraining to posttraining, and performance regressed substantially after the detraining phase. The BIL group produced a different response, with improvements at midtraining manifested in the BIL tests only. However, the peak power and jumping ability for the BIL group continued to increase, with significant improvements in all tests posttraining and, most importantly, did not regress during the detraining period.

These data tentatively indicate that although both UNI and BIL training are effective at increasing strength and power in the untrained, improvements from UNI may be more immediate, but shorter lasting. This training style seems most suited for situations that require rapid improvement (e.g., peaking for a competition or competing on short notice, etc.). However, its effectiveness on these grounds may peak after 6–8 weeks. BIL training may not provide similar initial gains in UNI power, but the adaptations seem to equate over time and last longer (at least up to 16 weeks). This is a critical factor in the maintenance of performance during periods of limited or no training (e.g., in-season training, recovery from injury, etc.). In reality, best practice probably warrants inclusion of both UNI and BIL exercises into a macrocycle, especially considering the minimum amount of information available at this point. Future information could very well alter the current conclusions.

CONCLUSION

Force production per limb often decreases during BIL contractions of homologous limbs (2,12–14,21–24,28,34,36,39,40). This BLFD is present in various populations (2,12,13,21–24,28,34,36,40). The magnitude of it is influenced by the training history of the person (14,28,35) and the exercise choice (e.g., multi vs. single-joint) (17,26,39), intensity (13,21), and velocity (40); however, age does not seem to be a significant factor. The BLFD also seems to be a dynamic and adaptable

neural phenomenon as chronic BIL training (17,23,37,38) tends to reduce it, whereas chronic UNI training may increase it (37,38).

Most importantly, the relationship between the BLFD and physical performance is almost entirely unknown. The only data available suggest that decreasing the BLFD through chronic BIL training increases physical performance (3). However, continual UNI and BIL training increase strength and power to a similar extent (17,29,32), with specificity of movement yielding the most benefit (9,29,38). For unknown reasons, chronic UNI training may increase performance more quickly, but BIL-induced adaptations allow greater long-term improvements and offer a defense against a loss in performance during detraining (27).

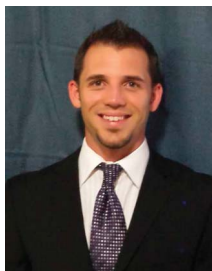
PRACTICAL APPLICATIONS

When taken together, the current information suggests that the BLFD should only influence exercise selection to a limited extent (if at all). In other words, the marker for the effectiveness of UNI and BIL exercises should be how they specifically alter physical performance, not how they influence the BLFD. Both UNI and BIL exercises are needed within a macrocycle to allow for both rapid increases and long-term development as competition nears. It is also important to consider that many other factors (such as the needs of the athlete, training history, equipment availability, desired training stimulus (25), and/or injury status) influence exercise choice. For example, given times of an injured limb (e.g., broken arm), UNI training of the healthy limb has been found to produce favorable adaptations of the immobile limb (5,7). Clinicians, therapists, and coaches should use their experience and evidence-based decision making to determine exercise selection, independent of the BLFD.

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Ramsey Nijem is a Masters Student at the Center for Sport Performance in the Department of Kinesiology at California State University.



Andrew J. Galpin is an Assistant Professor at the Center for Sport Performance in the Department of Kinesiology at California State University.

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